

4	(a) acceleration / force proportional to displacement (from a fixed point) <i>either</i> acceleration and displacement in opposite directions <i>or</i> acceleration always directed towards a fixed point	M1	A1 [2]
	(b) (i) g and r are constant so a is proportional to x negative sign shows a and x are in opposite directions	B1	B1 [2]
	(ii) $\omega^2 = g/r$ <u>and</u> $\omega = 2\pi/T$ $\omega^2 = 9.8/0.28$ $= 35$	C1	C1
	$T = 2\pi/\sqrt{35} = 1.06 \text{ s}$		A1 [3]
	time interval $\tau = 0.53 \text{ s}$		
	(c) sketch: time period constant (or increases very slightly) drawn line always 'inside' given loops successive decrease in peak height	M1	A1 A1 [3]